

Principal Component Analysis. Learning outcomes

- Carry out a principal components analysis using R.
- Assess how many principal components should be considered in an analysis (a given percentage of variance explained, where you see an *elbow* in the scree plot, components with eigenvalues greater than 1, etc.).
- Interpret principal component scores. This is usually done with the eigenvectors associated to each principal component, but you can also analyse the correlation between each principal component and each variable (`object_pca$var$cor` or with the command `dimdesc()`). Simply find which variables are most strongly correlated with each component, i.e., which of these numbers are large in magnitude. Which numbers we consider to be large or small is of course a subjective decision. You need to determine at what level the correlation value will be of importance and this will probably depend on the objectives, scientific foundation and questions of your study in real situations. Here, for instance, we might consider a correlation value above 0.5 or 0.6 (in absolute value) as important.
- Be able to describe a subject with a high or low score.
- Determine when a principal component analysis may be based on the variance-covariance matrix, and when the correlation matrix should be used. Principal component analysis should only be used with the raw data if all variables have the same units of measure. And even in this case, only if you wish to give those variables which have higher variances more weight in the analysis. If the variances of the variables are very different in magnitude, those with the highest one will tend to be emphasized in the first few principal components. Using the correlation matrix instead would prevent this from happening.
- Understand how principal component scores may be used in further analyses. Principal components are often treated as dependent variables for regression and analysis of variance.

1. To standardize or not?

- A key issue has to do with the scale of the variables.
- If variables have very different units of measurement, then we should standardize them to avoid variables with larger scales dominate the analysis.
- If variables have the same units you could leave them unstandardized or you could standardize the (recommended). You then compute the PCs using the matrix of correlations instead of the matrix of covariances.
- Regardless of the scaling decision, the data is always centered.

2. Interpreting PCA output (package FactoMiner)

Individuals or observations: When studying the individuals, we typically pay attention to:

- Scatterplots of PCs.
- Quality of the representation.
- Individual contributions to PCs.

The quality of representation of an observation on each axis (on a given dimension) is measured by the cosine-squared (`cos2`): `cos2` equal to 1 indicates that individual i is exactly on axis k (dimension k). Conversely, a `cos2` equal to 0 indicates that the individual i is on an orthogonal direction to axis k . Thus, the closer to 1 the better that observation is projected on that axis, the better it is represented by this component. It can help us to find components that are important to interpret that observation. Adding the squared cosines over all principal axes for a given observation we get 1. This has to do with the fact that all axes are needed in order to have the exact position of a point in the entire space.

$$\text{sum}(\text{object_pca}\$ind\$cos2[i,]) = 1$$

You can add the squared cosines of an individual over different axes, resulting in a *quality* measure of how well that individual is represented in that subspace.

The score or coordinate of an observation (when projected) on each component is given by `Dim.k`, where k refers to the component. These are the coordinates for the points on the observations' map and they are stored in the variable `object_pca$ind$coord`. The contribution (`ctr`) of observation i to component k is given as a percentage and stored in the variable `object_pca$ind$contrib`, and it is equal to:

$$\text{ctr}(i, k) = \frac{\text{coord}_{ik}^2}{\sum_{i=1}^n \text{coord}_{ik}^2} \times 100$$

Some remarks about contributions follow:

- `ctr( $i, k$ )` is the contribution of observation i to the construction of component k .
- The contribution is calculated as the percentage of inertia (variance) explained by the individual i on component k .
- The value of a contribution is between 0 and 100.
- For a given component, the sum of the contributions of all observations is equal to 100.
- The larger the value of the contribution, the more the individual contributes to the component.
- A useful heuristic is to examine those contributions that are larger than the average contribution, larger than $100/n$ (or $100/(n-1)$). If all individuals had a uniform contribution to a given PC, then they would have a contribution of $100/n$ (or $100/(n-1)$).

- Outliers have an influence and it is interesting to know to what extent their influence affects the construction of the components.
- Detecting those individuals that contribute to a given PC helps to assess the component's stability.

Dist is the length of the vector from 0 to that observation in the space of all principal components. For observation i (p is the number of variables and also the number of PC's if the data matrix has rank equal to p and $n > p$):

$$Dist = \sqrt{\sum_{k=1}^p coord_{ik}^2}$$

Thus, it is the distance of each observation to the center of gravity (the origin). This distance allows us to find which observations are *typical*, the closest ones to the center of gravity. Likewise, it allows us to identify the *unusual* or *unique* observations, those that have a large distance to the center of gravity. In general, the distance to the center of gravity is a criterion of the singularity of each observation or individual.

Variables: The focus is on the variables and their spatial properties: the angles between each pair of variables have cosines approximately equal to their pairwise correlations and this approximation is improved when the variables are well explained by the components, that is, close to the circumference of the unit circle. The coordinates for each variable in the circle of correlations are the correlations between each variable and the component (`object_pca$var$coord` = `object_pca$var$cor`). This correlation estimates the information they share and is collinear with the eigenvectors. FactoMiner provides the eigenvectors as part of the singular value decomposition, `object_pca$svd$V`. A high correlation between a component and a variable implies that the configuration of the individuals on that component resembles the position of the individuals on that variable. A unit correlation would indicate that the principal component is a linear function of the variable. A correlation close to 0 indicates there is no linear association between the principal component and the variable.

If the PCA analysis is done on the correlation matrix (normalized PCA), the cosine squared (**cos2**) is the squared of these coordinates:

$$object_pca\$var\$cos2 = object_pca\$var\$coord^2$$

A measure of the quality of each variable's representation by the first two principal components is given by:

$$cos2_{j1} + cos2_{j2}, \quad \text{for variable } j$$

The squared correlations (or **cos2**) are easier to interpret than the eigenvectors. They give the proportion of variance of each variable explained by each component and their sum, for a fixed variable over all components is equal to 1.

The contribution (ctr) of variable j to component k is given as a percentage (p is the number of variables):

$$ctr(j, k) = \frac{coord_{jk}^2}{\sum_{j=1}^p coord_{jk}^2} \times 100$$

The sum of all the contributions to an axis (to a component) is equal to 100 (as a percentage):

$$\sum_{j=1}^p ctr(j, k) = 100$$

Observation: Due to the fact that PCs are linear combinations of the original variables, they can be sometimes interpreted. Analysts try to label them in some meaningful ways. Unfortunately, although useful, this is not always possible.

3. Some more functions

Besides all the plots highlighting different aspects of observations and variables, other interesting functions are:

- *dimdesc()*
- *reconst()*
- *coeffRV()*. This coefficient measures the closeness of two sets of points, in this case, the proximity between two matrices. The null hypothesis is that this coefficient is equal to 0 and then the matrices aren't proximate. If it is equal to 0, each variable in the first matrix is uncorrelated to each variable in the second matrix. Here, a very small p-value means that the approximation is not so bad.